

TMS Architectural Design Session

Intelligence & AI-Driven Training Management System

Session Log

SYSTEM/ARCHITECT

Decision: Polyrepo approach chosen for the Training Management System (TMS).
Breakdown into Django (Core), FastAPI (AI), and Vue.js (Frontend).

USER

Generate a mmd diagram of the various entities in the system.

ASSISTANT

ERD generated covering Identity, Skill Taxonomy, Learning Tracks, and Governance.

USER

Create data entity objects of all entities in the system as json file.

ASSISTANT

JSON schema and entity definitions created for Identity, Org Structure, Skills, and Evaluation.

USER

Generate a mmd diagram of the features of admin command center.

ASSISTANT

Architectural feature diagram created for Admin Command Centre covering Data, Governance, Learning, and AI clusters.

USER

Create the mmd diagram of learner actions...

ASSISTANT

Learner action flow diagram created covering Discovery, Lifecycle, Analytics, and Engagement.

USER

Generate mmd diagram of manager's view...

ASSISTANT

Managerial view diagram created covering subordinates, badges, and organizational course tracking.

USER

Generate a mmd diagram of HR manager view...

ASSISTANT

HR Manager/RMT view diagram created for aggregate and granular resource views.

USER

In admin command center add feature to map trainers for various skills. Many to many mapping.

ASSISTANT

Feature Architecture updated to include Many-to-Many Trainer-Skill Mapping.

USER

As a software architect, help with the component design of the application.

ASSISTANT

Component design provided covering API Gateway, Core Governance (Django), and Intelligence (FastAPI) layers.

USER

Create a mmd file of the high level design diagram.

ASSISTANT

HLD Mermaid file created showcasing system layers and polyglot persistence.

USER

Create data flow diagram as .mmd file.

ASSISTANT

DFD Mermaid file created highlighting data ingestion, AI loops, and reporting outputs.

USER

What architecture design patterns would be useful...

ASSISTANT

Architectural patterns recommended: CQRS, EDA, Sidecar, Strategy, and Saga with rationale for each.

USER

Export the output as json.

ASSISTANT

Architectural design patterns and rationale exported to JSON file.